

Ben Fankhauser - Engineering Portfolio

L1 High-Power Certification Rocket

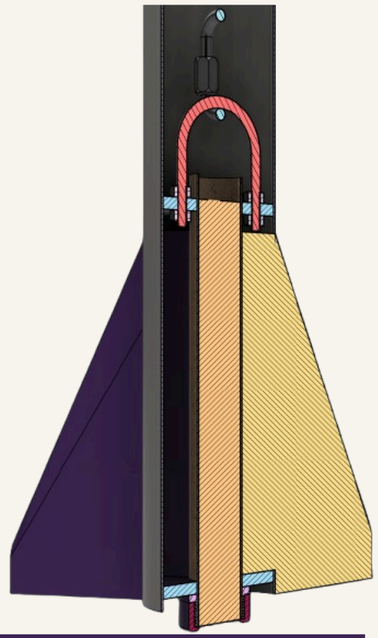
Sole Designer and Fabricator

Overview & Methods

- Project Description: End-to-end design and fabrication of a **3", 50" tall high-power rocket** for Tripoli L1 certification.
- Tools & Methods Utilized: OpenRocket simulation, custom CAD modeling, FDM 3D printing (PETG), laser cutting, precision fit-up, structural bonding.

Key design decisions

- Challenges Faced: Adding actual recovery hardware masses dropped simulated stability to under **0.9 cal**, below the safety threshold.
- Results and Outcomes: Redesigned fins, moving from 4 smaller fins to 3 larger fins with larger tip chord/span restored stability to **1.5+ cal**, but revealed an unsafe flutter margin **(0.9x)** at the original thickness. Bumping fin thickness to **3/16"** fixed flutter **(1.6x)** with negligible stability impact.
- Aerodynamic Profile Tooling: Designing & 3D-printed a custom sanding fixture to transform flat laser-cut fin stock into a 1/4-chord tapered airfoil profile, reducing pressure drag without compromising structural rigidity.

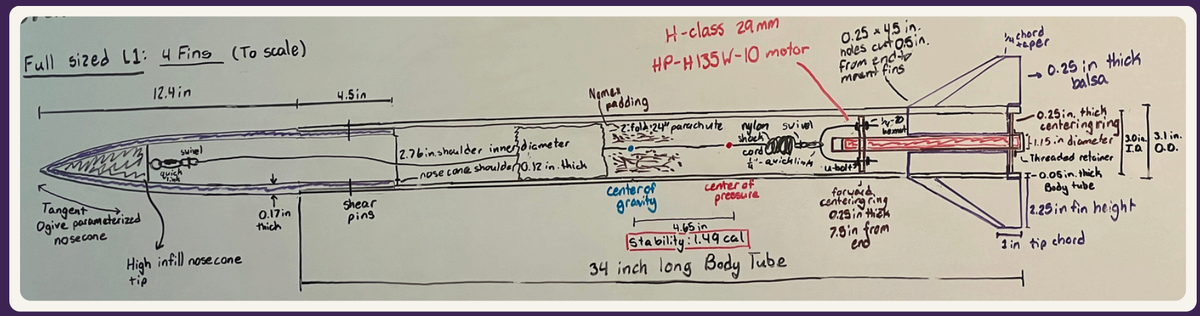


Open Rocket Simulation Data

Apogee (m)	Max. velocity (m/s)	Time to apogee (s)
920.7	194.5	12.2

Skills Developed

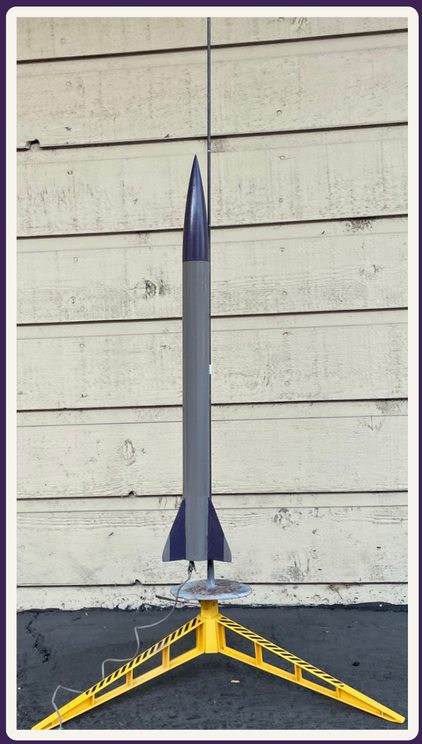
- Aerodynamic & Flutter Analysis: Evaluating Cp/Cg margins and calculating dynamic flutter boundaries under velocity constraints.
- Design-for-Manufacturing (DFM): Calibrating dimensional tolerances (contour, hole, first-layer compensation) for high-precision 3D-printed/laser-cut assemblies.



Status

Subscale test rocket built and to be flown soon for technique validation. Components for full scale arrived and to be built soon.

Initially planned to be built and flown September, 2026 but subscale testing faced delay due to the Washington state burn ban.



Red Ryder
Washington Aerospace Club

Manufacturing & Propellant Team

Overview

Contributing member on Red Ryder, a high-power rocket built by the Washington Aerospace Club post design phase. Contributed hands on building across AV bay assembly and decisions, motor mounting, fin alignment, carbon fin layup, shear pin hole drilling, and general assembly. One of three members who remixed the team's solid propellant grain after an earlier batch failed static-fire testing.



Contributions



- **AV Bay Assembly:** Cut out and assembled bulk heads, with threaded inserts to soundly attach through rods holding the AV slide rails.
- **General Fabrication:** Contributed to full assembly across multiple build sessions, doing epoxy and general manufacturing work based on drawings and specs.
- **Propellant Remix:** Worked with the WAC propulsion division WARP to remix the team's solid propellant grain to the original design recipe after an earlier batch was mixed incorrectly and failed static testing. Executed in strict protocol order to minimize friction combustion risks.





Skills Developed

Precision hand fabrication to spec, solid-propellant mixing procedure, working under strict safety guidelines and mixing timelines, working with existing engineering drawings rather than from scratch design.

South Sound Racing - UW Tacoma FSAE

Chassis Subteam

Reached out directly to UW Tacoma's newly formed Formula SAE club after connecting with the co-presidents at a Green River event to gain hands-on FSAE experience ahead of applying for the UW Seattle team.

Chassis design work

Working with club co-president to design early spaceframe chassis concepts to adhere to the FSAE 2026 Rulebook and align with other teams' design decisions.

Design Input:

Contributing early stage input on drivetrain and cooling/intake packaging during system layout.

Initiative

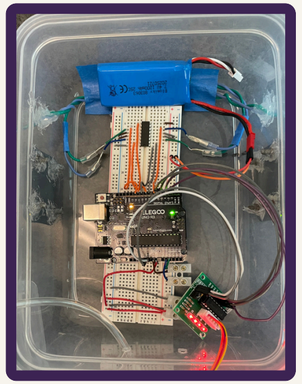
Independently sourced team access to an 800x800x1000mm 3D printer through external personal connection. Built and managing a print request system, handling all of the printing and coordination.

Skills Developed: Resource scheduling/access management, stakeholder coordination, working within strict competition guidelines.

Active-Ballast Submarine System

Independent Project, 2 person team

Overview



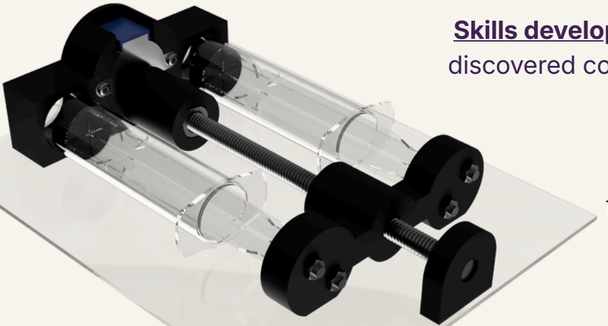
Began as a Green River Engineering Club ROV project: a basic container housing motors and controls. Once the quarter ended and the project concluded, my partner and I decided to continue the work and expand it into a true submarine, with me designing the stepper-driven ballast control system and handling all physical wiring and electronics assembly, and my partner handling circuit design and control code.

Key design decisions

- **Mechanism pivot:**
  - Built a single-syringe rack and pinion system first, found it wasn't self-locking, which risked water pressure back-driving the pinion when unpowered.
  - Realized two syringes were needed for balanced ballast. Driving both from one motor via rack-and-pinion would add gearing complexity, so switched to a self-locking leadscrew instead, driving both syringes symmetrically with no extra gear stage.

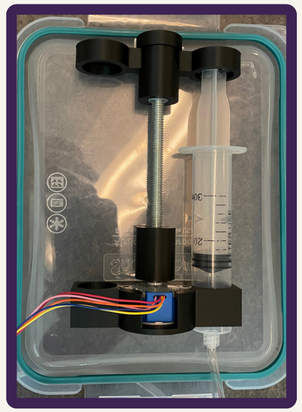
- **Motor speed fix:**
  - initial thread pitch gave ~3-4 min full ballast travel which was decided to be too slow for demo. Switched to a coarser thread pitch, cutting travel to ~2 min while keeping sufficient self-locking margin.

- **Manufacturing tolerance:**
  - used shrinkage-compensation methodology press-fit features held at true measured dimensions, clearance features offset to compensate for calibrated ~0.125mm undersizing.



Skills developed: tradeoff analysis, iterative redesign under a discovered constraint, shrinkage compensation for FDM parts.

Status: mechanical assembly complete and test-fit validated. Electronics integration confirmed with project partner, meeting soon for assembly and final sealing tests.



Wind Tunnel Instrumentation Concept

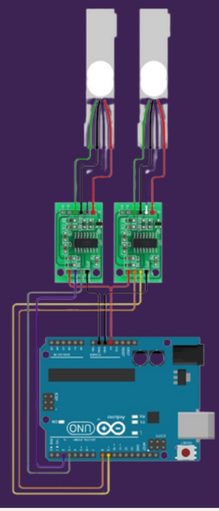
Independent Project

Designed a dual load-cell measurement concept to independently log downforce and drag on scale-model wind tunnel tests.

Concept: separate HX711 load-cell channels for drag and downforce, rather than one combined reading

Status: load cells and amplifiers sourced. Learning MATLAB/Simulink for signal analysis, not yet built or tested, still in research phase.

Skills Developed: sensor selection, load-cell signal conditioning, early MATLAB/Simulink and wind tunnel structural concept exposure through classes and independent research.



Trebuchet

Independent Project, Open Ended Physics Coursework

Self-designed and fabricated from scratch to test how counterweight mass affects launch range, later modeled in Fusion 360 for practice.

Data collection: recorded launches against a measured grid backdrop; calculated distance per frame, then converted to velocity using frame rate.

Result: across 4 counterweight masses **(501g–726g)**, average range increased from **210cm to 337cm**; measured range consistently ran **6–17%** below kinematic prediction likely due to drag and friction.

Skills developed: hand-tool fabrication, video-based data analysis, parametric CAD modeling

